3. X-Ray CT

1 Total variation:

$$f, g : \Omega \rightarrow R$$

- pointwise triangle inequality:

$$|\nabla (f+g)| = |\nabla f + \Delta g| \leq |\nabla f| + |\nabla g|$$

integrating over Ω:

$$TV(f+g) \leq TV(f) + TV(g)$$

subaditivity

$$\xrightarrow{c\,\nabla(cf) = c\nabla f} TV(cf) = \int_\Omega |c\nabla f|\,dx = |c|\,TV(f)$$

non-zero constant → zero gradiant
→ hence total variation

and TV is a seminorm → So, it is convex for $0 \leq t \leq 1$.

$$TV(tf + (1-t)g) \leq TV(tf) + TV((1-t)g)$$

$$= t\,TV(f) + (1-t)\,TV(g).$$ ✓

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2. Adjoint of the forward difference:

discrete inner product:

$$\langle f, q \rangle_h = h^2 \sum_{i,j} f_{i,j}\, q_{i,j}$$

extend the function by zero outside their bounded support

$$\langle D_x^+ f, P_1 \rangle = h \sum_{i,j} (f_{i+1,j} - f_{i,j}) P_{1,i,j}$$

$$\xrightarrow[i = i+1]{\text{reindex sum}} h \sum_{i,j} f_{i,j} P_{1,i-1,j} - h \sum_{i,j} f_{i,j} P_{1,i,j}$$

$$= -h^2 \sum_{i,j} f_{i,j} \frac{P_{1,i,j} - P_{1,i-1,j}}{h}$$

$$= \langle f, -D_x^- P_1 \rangle_h$$ ✓

$\Longrightarrow$ Same for y.

$$\nabla^* P = -(D_x^- P_1 + D_y^- P_2)$$ ✓

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3. Hurber function

$$S \geq 0 ; S = 0 \longleftrightarrow x = y = 0$$

$$\phi_\varepsilon(s) = \frac{x^2+y^2}{2\varepsilon}, \quad \frac{\partial \phi_\varepsilon(s)}{\partial x} = \frac{x}{\varepsilon}, \quad \frac{\partial \phi_\varepsilon(s)}{\partial y} = \frac{y}{\varepsilon}$$

$$\xrightarrow[S-\varepsilon/2]{S>\varepsilon} \frac{\partial \phi_\varepsilon(x)}{\partial x} = \frac{x}{S}, \quad \frac{\partial \phi_\varepsilon(y)}{\partial y} = \frac{y}{S}$$

if $S = 0$ then $x = y = 0$. From $\phi_\varepsilon(s) = \frac{(x^2+y^2)}{2\varepsilon}$,

both partial derivatives are 0.

$$\nabla_{(x,y)} \phi_\varepsilon(s) = \frac{(x,y)}{\max(\varepsilon, s)}$$

if $\nabla f = 0 \rightarrow |\nabla f|$ is nondiffrentiable.

So, the Hurber replacement is countiniuosly differentiable that means it is smooth.